



Figure 2: Gartner's Magic Quadrant for Enterprise Mobility Management, June 2015

moving away from the tech-centric position of the past towards a more business-aligned approach, expanding support for the app development life cycle and offering more robust and scalable capabilities.

Depending on an enterprise's needs, a vendor in any quadrant could be the best for an enterprise. Some enterprises may prefer custom MADP to address specific concerns related to vendor lock-in, the nature of the solution, security, roadmap demands, long-term costs, etc.

Enterprise Mobility Management (EMM)

Gartner has defined the EMM space as: "Enterprise mobility management (EMM) suites consist of policy and configuration management tools and a management overlay for applications and content intended for mobile devices based on smartphone OSs. They are an evolution from the previous generation of MDM (mobile device management) products that lacked application and content management. IT organisations and service providers use EMM suites to deliver IT support to mobile end users and to maintain security policies."

Broadly speaking, there are three major categories of EMM solutions, namely: mobile device management (MDM), mobile application management (MAM) and mobile content management (MCM).

An MDM solution provides for device level management functionality. For instance, a company may want to restrict an employee to using a corporate-owned device to connect to a network through the company Wi-Fi only. The features of an MDM solution are heavily dependent on the support provided by the target device platforms. The more manageability a platform provides, the better the control that an MDM solution will have on those devices.

Some policies typically used under MDM are passcode enablement, Wi-Fi restriction, application whitelisting, mandated VPN, storage encryption, etc.

MAM focuses on securing corporate data from possible unauthorised access and misuse. Typically, an application management solution supports app level policies like blocking screen capture, copying of screen data, offline data encryption, etc, to ensure that the corporate data remains sandboxed and protected.

MCM takes care of securing the corporate content accessed and shared on the devices in the form of files. It provides the users a way to access, store and view documents from enterprise content repositories while letting the administrator establish controls to protect this content from unauthorised and unsafe distribution.

Some controls usually deployed under this category include document encryption, access control rules on sharing, a file sync policy, etc.

Some commercial EMM suites even provide for add-ons related to network management — for example, secure apps for browser and personal information management.

You can see the Gartner Magic Quadrant for Enterprise Mobility Management 2015 in Figure 2. There are many EMM solutions out there, ranging from cloud-based to site-based deployment options. While the SaaS based model is becoming the preferred choice in the industry, the EMM focal point is gradually shifting to data and unified endpoint management.

Surely, EMM is becoming necessary for any organisation that manages more than a handful of mobile devices. Assessing your company's needs and accordingly choosing the right EMM solution is the key to achieving the expected return on investment (ROI).

Building mobile applications for enterprises requires a different level of understanding and work, as compared to consumer mobile applications. The enterprise mobile solution not only has to stand out in terms of user experience, but also has to integrate well with the existing enterprise set-up, and needs to ensure security and manageability of the data leaving the enterprise premises.

Some of the recommended open source tools and technologies that can be leveraged for building robust enterprise mobile solutions are: OpenStack for IaaS, Cordova for cross-platform development, Robotium and Selenium for test automation, Robolectric and Mockito for mobile app unit testing, etc.

By leveraging the recommended solutions, an enterprise can achieve both secure and manageable solutions that are capable of transforming how its business processes work, in a time-efficient and cost-effective manner. 

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